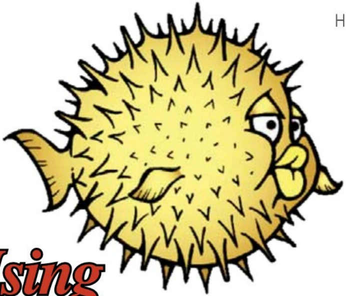

Using OpenBSD for the Server Infrastructure

Some regular readers of have probably used Linux as a desktop, while others may have even done programming on it. This article gives readers an overview on using OpenBSD, a UNIX operating system. Read on to find out how this amazing OS can be put to good use.

You can download this OS for free from *www.openbsd.org* and install it on a laptop or desktop, using the CD ISO image. Installing OpenBSD is not hard, and may involve some trial and error.

OpenBSD is a very clean OS, with a lot of great ideas and discipline having gone into its design and coding conventions. It is slim, neat, clean and elegant in every way. It lacks support for a few of the fancy and frivolous things that some end users may think necessary, but appears to meet the goals of most scientists and engineers. OpenBSD, though a general-purpose OS as UNIX has always been, is an out-and-out developer's OS or engineer's friend.

The differences between OpenBSD and Linux

Linux comprises the kernel (begun by Linus Torvalds) plus the GNU userland, Bash shell, GNU compiler suite, applications and glibc. The projects involved are different, maintained separately and follow different timelines and release schedules, making it hard for end users to do a combination of these themselves—so various distributions and vendors like Debian, Red Hat, Slackware or Ubuntu blend these elements to make end user-viable operating systems. Also, obtaining the source code for every part of the OS is difficult, since it is not a single project.

In contrast, for OpenBSD, the entire OS, the base system, the kernel, the userland, the packages and the glue code -- all go into the same project. They are developed and maintained by the same team. A particular release of OpenBSD (say, 5.1) is unique, and depending on the hardware architecture you can get the exact set of binaries and source. (This is also true of NetBSD and FreeBSD.)

My tryst with OpenBSD

I have been using OpenBSD for my personal and commercial activities since 2003. I am a cryptographer, and I had to develop the IPsec kernel crypto code for my former employer's router device. I looked at the IPsec implementation in FreeSWAN, the KAME project in FreeBSD, and also looked at NetBSD and OpenBSD. I finally settled on OpenBSD, and ended up working in the kernel C code. That is how I started my journey—and I have never looked back since. For this article, I will only cover IT infrastructure usage, using OpenBSD as a server and desktop OS, and as a developer platform. I will not be too concerned with kernel or C coding and development (other than once in a while). By and large, I will cover real-world personal and business use of this platform.

I have created my own USB installer, and I have my own USB release of OpenBSD, which I maintain in eight separate projects. You can access it at *http://liveusb-openbsd.sf.net.*